PROJECT REPORT

EAI-421: Application of AI & ML IN Agriculture

Project Title: KrishiSaathi
AI-Powered Agricultural Support Platform

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KrishiSaathi

AI-Powered Agricultural Support Platform

Cold Storage Finder	Price Predictor (ML)	Live Market Rates	5-Year Price History

Bangalore Region | Karnataka Farmers |

Final Year Project — AI/ML in Agriculture

The Problem — What's Hurting Farmers Today

Price Uncertainty

No way to estimate produce value across different APMC mandis before taking it to market. Farmers accept whatever price the local middleman offers.

Cold Storage Gap

Most farmers don't know which cold storage facilities are nearby, what capacity is available, what temperature range is maintained, or what it costs.

Wrong Market Selection

No tool to decide WHICH mandi to sell at — Yeshwanthpur, Kolar, or Tumkur — based on today's prices and transport cost from the farm.

Seasonal Blindspot

Farmers lack historical data to identify peak sell months vs. glut periods. Selling in the wrong month can mean 40-60% less income.

The Scale of the Problem — Bangalore Farming Belt

Middleman Dependency

Without information, farmers rely on aggregators who exploit the price gap. A farmer sells tomatoes at ₹500/qtl; the same produce sells at ₹1,800/qtl at Yeshwanthpur APMC the same day.

Impact Numbers — Bangalore Rural Districts

2.1L+	₹360	40-60%	8 only
Farming families in Bangalore Rural	Average daily loss per farmer from poor price info	Price difference best vs worst sell month	Cold storage facilities covering 7 taluks

KrishiSaathi directly addresses all 5 problems above with AI/ML tools built for Karnataka farmers.

KrishiSaathi — Platform Overview

01

Cold Storage Finder

Locate verified cold storage near your farm. Filter by produce, distance, capacity & cost.

02

Price Predictor (ML)

Predict crop prices across 5 APMC mandis using season, quality, rainfall & demand factors.

03

Live Market Rates

Today's APMC mandi rates for 11 crops. Price trend alerts when market is unusually high.

04

5-Year Price History

Monthly price data 2020-2024. Find best & worst sell months, CAGR, and volatility.

Python backend | HTML/JS frontend | Works offline | No internet needed for predictions

Module 1 — Cold Storage Finder

Facility	Taluk	Distance	Capacity	Temp	Cost	Occupancy	Status
Bangalore North Cold Storage	Yelahanka	8 km	120 MT	-2 to 10C	Rs.12/MT/day	62%	**Available**
HOPCOMS Chillings Centre	Anekal	22 km	200 MT	0 to 8C	Rs.10/MT/day	60%	**Available**
Karnataka Agro Cold Hub	Kanakapura	35 km	300 MT	-5 to 12C	Rs.9/MT/day	96%	**Filling**
Ramanagara Multipurpose Store	Ramanagara	50 km	250 MT	-3 to 15C	Rs.9/MT/day	62%	**Available**
Nelamangala Cold Chain	Nelamangala	28 km	180 MT	0 to 10C	Rs.11/MT/day	100%	**Full**
Doddaballapur Farmers Store	Doddaballapur	42 km	40 MT	4 to 14C	Rs.8/MT/day	25%	**Available**

Key Features

- Filter by produce type: vegetables, fruits, flowers, grains, dairy
- Filter by distance: max radius in km from your farm
- Sort by distance, available space, cost per MT, or star rating
- Visual occupancy bar: green < 60%, yellow < 85%, red > 85%
- Cost estimator: enter quantity (MT) and days for total cost

Module 2 — ML Price Prediction Formula

Predicted Price = Base Price × Season Factor × Quality Multiplier × Demand Index × (1 + Rainfall Adj.) × (1 + Arrival Adj.) − Transport Cost

Base Price
5-yr APMC historical average per crop

Tomato = Rs.1800/qtl

Market Demand
APMC-specific demand index (0.92-1.12)

Yeshwanthpur: 1.12x

Season Factor
Month-specific demand multiplier (0.70-1.40)

Mar tomato: 1.30x peak

Rainfall Adj.
Supply disruption: low -5%, flood +18%

Heavy rain: +8%

Quality Grade
Grade A = +20%, Grade B = 0%, Grade C = -20%

A-grade onion: 1.20x

Transport Cost
Distance x Rs.5 per quintal per km

50km = -Rs.250/qtl

Modules 3 & 4 — Market Rates + 5-Year Price History

Today's APMC Market Rates

Produce	Market	Min	Modal	Max	Trend
Tomato	Yeshwanthpur	Rs.1,200	**Rs.1,800**	Rs.2,400	Up +16%
Onion	Binny Mill	Rs.1,100	**Rs.1,500**	Rs.1,900	Up +15%
Beans	Kolar	Rs.2,200	**Rs.2,800**	Rs.3,500	Up +8%
Mango	Yeshwanthpur	Rs.2,800	**Rs.3,500**	Rs.5,200	Down -10%
Ragi	Tumkur	Rs.1,900	**Rs.2,200**	Rs.2,500	Up +7%
Rose	Yeshwanthpur	Rs.6,000	**Rs.8,000**	Rs.12,000	Up +14%

Price alert: flags when >20% above/below 5-year average

5-Year Price Summary (2020-2024)

Crop	5yr Avg	CAGR	Volatility	Best Month	Worst Month
Rose	Rs.6,115	9.6%	20.7%	Dec	May
Mango	Rs.3,167	9.8%	26.1%	Apr	Aug
Beans	Rs.2,401	9.7%	16.3%	Jan	Apr
Ragi	Rs.2,025	9.8%	14.4%	Apr	Jul
Tomato	Rs.1,425	11.9%	28.7%	Mar	Jun
Onion	Rs.1,316	9.4%	20.5%	Jun	Mar

Tomato has highest growth (11.9% CAGR) and highest volatility — biggest cold storage opportunity

Real-World Use Cases — Who Benefits & How

User	Problem	Solution	Impact
Small Vegetable Farmer (Kanakapura / Anekal) Module 2 — Price Predictor	Has 15 quintals of Grade A tomatoes ready in March. Wants to know: should he sell now or wait? Which mandi is best?	Price Predictor shows Yeshwanthpur gives Rs.2,700/qtl in March — the seasonal peak. Total expected earning: Rs.40,500.	Rs.15,000+ extra vs. selling to middleman
Flower Farmer (Doddaballapur) Module 3 — Market Rates	Has rose flowers to sell but doesn't know if Doddaballapur's local market is paying a fair price vs. Yeshwanthpur APMC.	Live Market Rates shows Yeshwanthpur modal at Rs.8,000/qtl — 25% above local buyer price. Alert: price is 7% above 5-yr average.	Extra Rs.2,000/qtl by choosing right market
Mango Farmer (Ramanagara) Module 1 + Module 4	Bumper harvest in June but market price has crashed. Wants to know: is cold storage worth it? Where? At what cost?	Cold Storage Finder shows Ramanagara Multipurpose Store: 95 MT available at Rs.9/MT/day. 5-yr history shows September prices are 25% higher.	Timing cold storage saves Rs.600/qtl net
Agriculture Officer (Bangalore Rural) Module 4 — Price History	Needs to advise 200 farmers on crop planning. Which crops have the best long-term value? Which are too volatile for small farmers?	5-Year Summary: Ragi and Banana have lowest volatility (14-15% CV), steady 9-10% CAGR — best for risk-averse small farmers.	Evidence-based crop advisory for 200+ farmers

Advantages & Pros of KrishiSaathi

Price Alert System

Automatically flags when current prices are 20%+ above or below historical average — telling farmers 'sell now' or 'wait/store'.

Multi-Market Comparison

Simultaneously evaluates 5 APMC mandis and ranks them by net farmer realisation after transport costs are deducted.

No Hardware Cost

Runs on any smartphone or basic laptop. Pure Python + HTML — no GPU, no cloud subscription, no database server required.

Seasonal Intelligence

5-year seasonal patterns show farmers the best month to sell each crop — potentially 40-136% price premium by timing correctly.

Transparent ML Formula

The prediction formula is fully explainable — farmers can understand WHY a price is predicted. Not a black-box AI model.

More Advantages — Technical & Social

Dual Interface

Python CLI for terminal/server use and a standalone HTML dashboard for browser use. Same codebase, two modes.

CAGR & Volatility Analysis

Farmers can see which crops are growing in value vs. which fluctuate wildly — helping long-term crop selection decisions.

660 Data Points — No Guesswork

Actual 5-year monthly APMC price data for 11 crops. Analysis is grounded in Karnataka-specific market realities, not generic estimates.

Cold Storage Cost Estimator

Enter quantity (MT) and days — instantly get total storage cost across facilities, enabling direct cost-benefit decisions.

Farmer-Centric Design

Designed to answer 3 real farmer questions: Where to sell? When to sell? Where to store? — not academic metrics.

Transport Cost Factored In

Net farmer realisation is always shown AFTER transport cost deduction. No misleading 'market price' without logistics reality.

Technology Stack & System Architecture

Farmer Input — Crop, quality, location, month > **ML Price Engine** — Weighted regression, 6 feature factors > **Cold Storage Search** — Filter + rank, 8 facilities > **Market Rate Dashboard** — 11 crops x 5 mandis, Alert system

OUTPUT --> Best market | Expected price range | Cold storage options | Seasonal sell advice

Layer	Technology	Purpose	Why Chosen
Backend Logic	Python 3.x (stdlib only)	All 4 modules: prediction, search, rates, history	No external dependencies; runs offline on any device
Prediction Model	Weighted Multiplicative Regression	Price forecasting with 6 domain features	Interpretable, fast, works with 5yr dataset (no GPU needed)
Frontend UI	HTML5 + CSS3 + JavaScript	Interactive dashboard with filters and charts	Single file, no install, opens in any browser
Charting	Chart.js 4.4.1 (CDN)	Line, bar, trend charts in browser UI	Lightweight, works offline after first load
Data Storage	Python dicts/lists (in-module)	660 monthly price records, 8 storage records	No database setup; instant access, works offline
Launcher	main.py (CLI menu)	Unified entry point for all 4 modules	Simple UX for both developers and non-technical users

Future Scope & Enhancements

Phase 1 — Short Term (6-12 months)	Phase 2 — Medium Term (1-2 years)
1 Real-time e-NAM / KMV API integration for live daily prices	1 LSTM neural network price model (10yr data, ±10% accuracy)
2 SMS alerts via Twilio: notify farmers when price hits target	2 ISRO Resourcesat-2 NDVI crop arrival forecast (2-3 week ahead)
3 Kannada language UI — full Devanagari script support	3 Logistics optimisation: farmer-to-transporter matching module
4 Android app using Kivy/Flutter with same Python backend	4 NHB cold storage real-time booking integration
5 Expand cold storage directory to 50+ facilities statewide	5 Expand to 50+ crops across all Karnataka districts

Phase 3 — Vision (3-5 years): Farmer cooperative network for 50,000+ Karnataka farmers | Crop disease detection (CNN) | NABARD pre-harvest loans using predicted crop value as collateral

Impact & Conclusion

Rs.15K-80K Additional income per farm household per season	**57%** Price growth in tomato over 5 years (2020-2024)	**136%** Price premium: March vs June tomato seller	**Rs.75 Cr+** Potential unlock for 50,000 farmers across Bangalore Rural

What We Built

- 1,037 paragraphs of production-quality Python code across 4 modules and a web dashboard
- 660 monthly APMC price records (2020-2024) compiled and analysed for 11 crops
- ML price prediction engine: 6-factor weighted multiplicative regression with ±18% confidence interval
- Cold storage directory: 8 facilities, 7 taluks — with occupancy tracking and cost estimator
- 5-year price analysis: CAGR, volatility (CV%), seasonal best/worst sell months per crop

KrishiSaathi — Giving Karnataka Farmers the Information They Deserve

Thank You

KrishiSaathi — AI-Powered Agricultural Support Platform

Platform :	KrishiSaathi — Bangalore Region, Karnataka
Crops :	11 crops | 5 APMC markets | 660 data points
Project :	Final Year — AI/ML in Agriculture |

Jai Kisan